

Varshini Elangovan

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EDUCATION

Carnegie Mellon University (CMU)

Master of Science in Machine Learning

Expected: December 2026

Pittsburgh, PA

Anna University, College of Engineering, Guindy

Bachelor of Engineering (Honors) in Computer Science and Engineering, CGPA: 9.82/10.0

June 2022

Chennai, India

WORK EXPERIENCE

Google

Software Engineer

Hyderabad, India

July 2022 – July 2025

- Primary contributor to the design of the risk strategy for Google Wallet via PIX in Brazil launched in August 2024. Developed risk mitigation solutions to reduce transaction losses across peer-to-peer payment flows.
- Designed and deployed feature engineering pipelines in Java and rule-based decision policies in Python, integrating ML model outputs for real-time transaction risk assessment at scale.
- Implemented regulation-compliant risk capabilities including real-time monitoring dashboards and operational runbooks. Collaborated with Trust & Safety and Product teams on metrics, thresholds and customer impact post-launch.
- Engineered ML pipelines for delinquency fraud detection on Stored Value transactions, covering data simulation, GBDT model training in TensorFlow and offline evaluation against baselines.
- Promoted from SWE II to SWE III (November 2024) for high-impact contributions to Global Payments Platform Risk systems.

Goldman Sachs

Summer Analyst

Remote (Bengaluru, India)

June 2021 – July 2021

- Created a Python-based data curation and analysis tool that converts JSON inputs into multiple outputs for GS proprietary systems. Integrated the same into ETL workflows, optimizing a 2-week process to under one minute.

RESEARCH EXPERIENCE

Human Sensing Lab, Carnegie Mellon University

Graduate Student Researcher

Pittsburgh, PA

October 2025 - Present

- Researching 3D Spatial Understanding with 2D Vision-Language models under Prof. Fernando De la Torre in a collaboration project with Meta Reality Labs. Developing methods to integrate semantic priors into 2D VLM inputs for spatiotemporal scene understanding and visual question answering.

Indian Institute of Technology (IIT), Kharagpur

Deep Learning Intern

Remote (Kharagpur, India)

May 2022 – September 2022

- Trained and evaluated semantic segmentation models in PyTorch for depth estimation from simulated colonoscopy images as part of the KLIV group, assessing various loss functions and effective post-processing techniques. Presented at the MICCAI 2022 Endoscopic Vision Challenge.

PROJECTS

Watch & Learn: Teacher-Student Distillation for Robotic Arm Tasks ([Report](#))

November 2025 – December 2025

- Developed a vision-only student agent from the distillation of a full-state teacher policy for camera-based robotic manipulation on MetaWorld tasks. Achieved 90% of full-state teacher performance via teacher-guided RL with imitation learning.

Let's Dance: Music-conditioned 2D Dance Movement Generation

January 2022 – June 2022

- Implemented an end-to-end GAN framework in PyTorch to synthesize dance motion videos based on audio, leveraging OpenPose 2D skeletal key points and a pix2pix model to generate temporally coherent photo-realistic motion sequences.

PUBLICATIONS

- SimCol3D — 3D reconstruction during colonoscopy challenge**, A. Rau, S. Bano, Y. Jin, Varshini Elangovan, et al., Medical Image Analysis, 2024. doi: [10.1016/j.media.2024.103195](https://doi.org/10.1016/j.media.2024.103195)
- Reverse Image Search Engine for Garment Industry**, Anushri Eswaran and Varshini E., 8th International Conference on Advanced Computing and Communication Systems (ICACCS), 2022, doi: [10.1109/ICACCS54159.2022.9785089](https://doi.org/10.1109/ICACCS54159.2022.9785089)

SKILLS

- Languages:** Python, Java, C/C++, SQL, JavaScript
- Frameworks/Libraries:** PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, OpenAI Gym
- Developer Tools:** Git, Docker